London NRMM LEZ

Compliance Effect Analysis

Data period: 2016–2024

n = 10,749 audit records across three equipment groups

Research Plan v1.18 | 31 March 2026

Executive Summary

Three-component decomposition of stage-based compliance improvement into natural market turnover, voluntary proactive LEZ response, and direct enforcement.

Principal findings

- **59.7%** of all audit records are self-compliant at initial audit — no enforcement needed
- **22.2%** are driven compliant via *registration* only — no stage change results
- Only **8.7%** involve a machine below the stage threshold that is removed or driven compliant — the records that produce actual emissions change
- Stage non-compliance rate: **13.8%** of warm fleet vs 29.6% from `Initial.Machinery.Compliance` — gap is entirely registration-only failures
- **Rest of London Phase B** stage $\bar{e}$ = 4.0% — the RoL fleet was already largely above IIIB threshold; its 28.1% `Initial.Machinery.Compliance` rate reflects unregistered machines, not emissions failures
- Voluntary proactive LEZ response (λ_Proactive) is detectable in all groups when estimation is restricted to self-compliant warm records

Data and Fleet Structure

Equipment groups

Group	Description	Phase B threshold
Constant Speed (CS)	Generators and constant-speed plant	Stage V (≥6)
CAZ+	Variable-speed, Central Activity Zone + Olympic Area	Stage IV (≥5)
Rest of London (RoL)	Variable-speed, Greater London outside CAZ+	Stage IIIB (≥4)

Stage mapping: I=1, II=2, IIIA=3, IIIB=4, IV=5, V=6, ZE=7

Model phases

Phase	Period	Key change
A1	1 Jan 2016 – 31 Dec 2018	Initial programme
A2	1 Jan 2019 – 31 Aug 2020	Stage V equipment newly available
B	**1 Sep 2020 – 31 Dec 2024**	**Threshold step-up: CS→V, CAZ+→IV, RoL→IIIB**

Phase B subdivided into three equal segments (~527 days) for variable-speed estimation.

Compliance Routing — Classification Rules

Every audit record assigned to one of eight outcomes using three fields:

`Initial.Stage` vs threshold · `Initial.Machinery.Compliance` · `Final.Machinery.Compliance`

Routing	Stage OK?	`Initial.Machinery.Compliance`	Final outcome	Stage change?
Self-compliant	✓	compliant	—	No
Compliant — exemption/viability	✕	compliant	—	No
Driven compliant — registration	✓	non-compliant	compliant	No
Removed — registration	✓	non-compliant	removed	No
Non-compliant — registration	✓	non-compliant	unresolved	No
Driven compliant — emissions	✕	**non-compliant**	**compliant**	**Yes**
Removed — emissions	✕	**non-compliant**	**removed**	**Yes**
Non-compliant — emissions	✕	non-compliant	unresolved	No

Only the two **bold** rows produce a stage transition through enforcement.

Compliance Routing — Overall Counts (n = 10,749)

Routing	n	%
Self-compliant	6,417	59.7%
Driven compliant — registration	2,388	22.2%
Removed — emissions	**557**	**5.2%**
Driven compliant — emissions	**380**	**3.5%**
Compliant — exemption/viability	374	3.5%
Non-compliant — registration	356	3.3%
Non-compliant — emissions	216	2.0%
Removed — registration	61	0.6%

Registration-driven routes together: **26.1%** of all records — none produce a stage change.

Emissions-driven enforcement (stage change): **8.7%** (937 records).

Compliance Routing — Year-by-Year

Cold fleet (top) · Warm fleet (middle) · All records (bottom)

Decomposition Framework

$$\lambda_{\text{Policy}} = \lambda_{\text{CF}} + \lambda_{\text{Proactive}}$$

Three components

Component	Symbol	Estimated from
Counterfactual rate	λ_CF	Cold-engaged fleet stage distribution
Policy rate	λ_Policy	Warm fleet, **self-compliant records only**
Proactive rate	λ_Proactive	Derived: λ_Policy – λ_CF (floored at 0)

Enforcement exposure

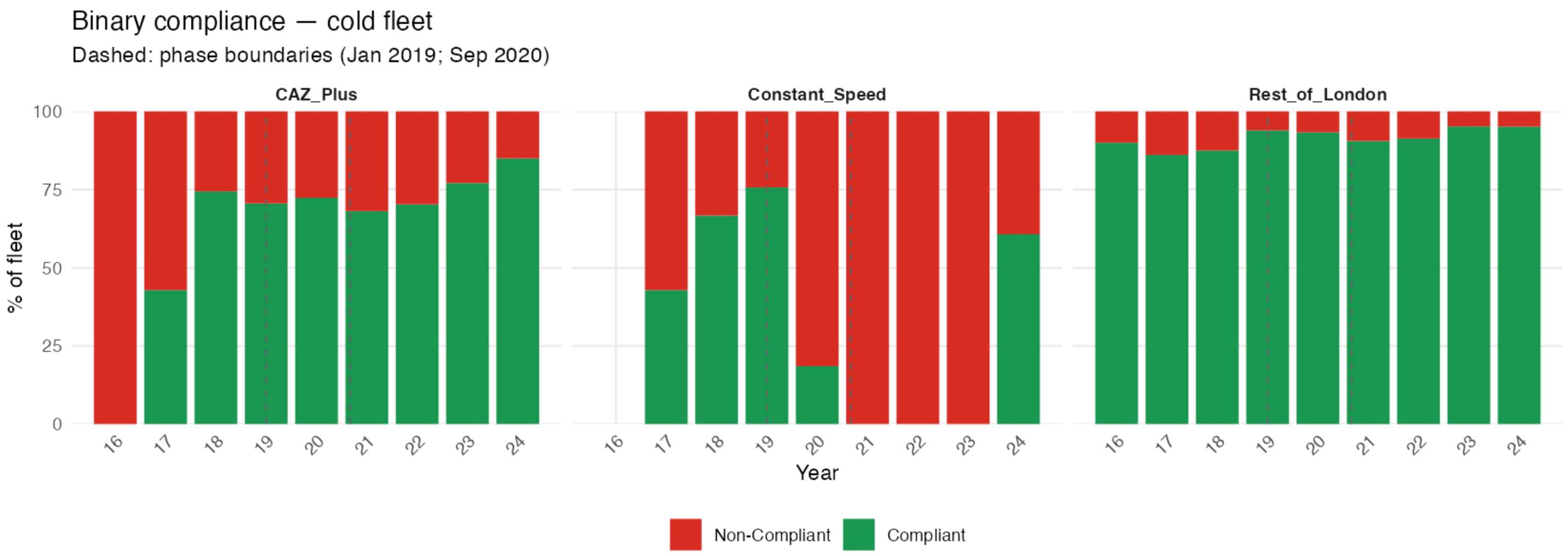
$$\bar{e}_{g,\phi} = \frac{\#\{records : \text{Initial.Stage} < \text{threshold}_{g,\phi}\}}{N_{g,\phi}^{all}}$$

Counts records where stage is below threshold across **all records** (cold + warm). Registration-only non-compliance excluded by construction.

WLS estimator

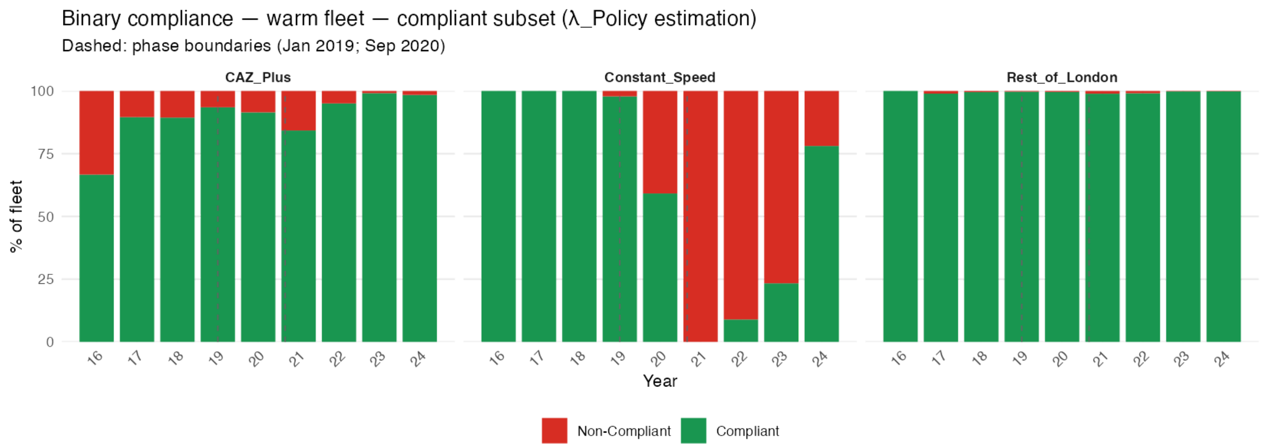
$$\frac{\Delta p_{c,t}}{\Delta t} = \lambda \cdot (1 - p_{c,t})$$

Cold Fleet — Compliance Profiles



Policy Rate (λ_Policy) — Self-Compliant Warm Subset

Estimated from warm records that are **self-compliant at initial audit** — machines meeting stage threshold, registered. Captures voluntary proactive LEZ response free from enforcement contamination.



λ_Policy estimates

Case	n	λ_Policy	SE	95% CI	Significant?
A1/A2 Variable Speed pooled	2,094	**0.361**	0.162	[0.043, 0.679]	Yes
CS Phase B	252	**0.264**	0.182	[−0.093, 0.620]	Marginal
CAZ+ Phase B	**1,416**	**0.536**	**0.024**	**[0.490, 0.582]**	**Yes — high precision**
RoL Phase B	2,298	**0.338**	0.177	[−0.010, 0.685]	Marginal

Enforcement Exposure (ē)

Group	Phase	N (all records)	N (Stage < threshold)	ē
CS	A1	104	23	0.221
CS	A2	316	56	0.177
CS	**B**	**692**	**492**	**0.711**
CAZ+	A1	400	48	0.120
CAZ+	A2	658	50	0.076
CAZ+	B	2,292	457	0.199
RoL	A1	538	26	0.048
RoL	A2	1,634	52	0.032
RoL	**B**	**4,115**	**162**	**0.039**

CS Phase B ē = **0.711** — the Stage V mandate left 71% of CS machines non-compliant on emissions.

RoL Phase B ē = **0.039** — RoL fleet was already largely above IIIB threshold.

Counterfactual Rate (λ_CF)

λ_CF = annual rate of stage improvement without LEZ intervention.

Estimated from the cold-engaged fleet (sites not registered with the LEZ).

Stream B granular estimates (authoritative)

Group	λ_CF	SE	95% CI
Constant Speed	**0.358**	0.067	[0.226, 0.490]
CAZ+	**0.363**	0.052	[0.261, 0.464]
Rest of London	**0.219**	0.048	[0.125, 0.313]

Bias direction

The cold fleet **overestimates** the true counterfactual rate — LEZ demand elevates second-hand stock entering unregistered sites. λ_CF is a conservative upper bound; λ_Proactive is a conservative lower bound.

CS Phase B: λ_CF = 0 confirmed by inspection — no Stage V machinery in cold CS fleet 2021–2023.

Enforcement Exposure — Stage vs `Initial.Machinery.Compliance`

Positive gap = `Initial.Machinery.Compliance` overcounts by including registration failures (no stage change).

Group	Phase	Stage ē (warm)	`IMC` ē (warm)	Gap
CAZ+	A2	0.095	0.282	+0.187
CAZ+	B	0.192	0.310	+0.118
CS	B	**0.828**	0.526	−0.302
RoL	A2	0.025	0.285	+0.260
RoL	**B**	**0.040**	**0.281**	**+0.241**

RoL Phase B: 24 percentage point gap = registration-only non-compliance with no emissions consequence.

CS Phase B negative gap: many CS machines hold viability exemptions — `Initial.Machinery.Compliance` = compliant despite Stage < V threshold.

Full Decomposition

$$\lambda_{\text{Proactive}} = \lambda_{\text{Policy}} - \lambda_{\text{CF}} \quad \text{(floored at 0)}$$

Case	λ_CF	λ_Policy	λ_Proactive	ē	λ_Pro / λ_Policy	Reliability
A1/A2 Var Speed	0.130	0.361	**0.231**	0.076	64%	Good
CS Phase B	0.000	0.264	**0.264**	0.711	100%	Moderate
CAZ+ Phase B	**0.161**	**0.536**	**0.375**	**0.199**	**70%**	**Excellent**
RoL Phase B	0.152	0.338	**0.186**	0.039	55%	Moderate

CS Phase B: λ_CF = 0 → every unit of voluntary compliance is entirely attributable to the proactive LEZ response.

CAZ+ Phase B: most precisely estimated case; 84% → 98.7% compliance in self-compliant warm subset over Phase B; proactive response (0.375/yr) is the largest of any group.

RoL Phase B: genuine but modest proactive response (0.186/yr) against very low stage-threshold enforcement exposure — policy effect here is predominantly through registration compliance, not stage upgrading.

Key Results Summary

Finding	Value
Self-compliant at initial audit (all records)	**59.7%** (6,417)
Driven compliant — registration (no stage change)	**22.2%** (2,388)
Driven compliant or removed — emissions (stage change)	**8.7%** (937)
Persistent non-compliant	**5.3%** (572)
Stage non-compliance rate — warm fleet	**13.8%**
λ_{CF} — Constant Speed	0.358/yr [0.226, 0.490]
λ_{CF} — CAZ+	0.363/yr [0.261, 0.464]
λ_{CF} — Rest of London	0.219/yr [0.125, 0.313]
λ_{Policy} — CAZ+ Phase B	**0.536/yr** [0.490, 0.582]
$\lambda_{Proactive}$ — CAZ+ Phase B	**0.375/yr** (70% of λ_{Policy})
$\lambda_{Proactive}$ — CS Phase B	**0.264/yr** (100%; $\lambda_{CF} = 0$)
$\bar{e}$ — CS Phase B (highest)	**0.711**
$\bar{e}$ — RoL Phase B (lowest)	**0.039**